Davenport–Schinzel Sequences: The Critical Regime

1 Definition

Definition 1 (Order-s Davenport–Schinzel sequence)**.** Let $n, s \geq 1$ and $[n] = \{1, \ldots, n\}$. A sequence $U = (u_1, \ldots, u_m)$ over $[n]$ is an order-s Davenport–Schinzel sequence if

1. $u_i \neq u_{i+1}$ for every i; and
2. for every two distinct symbols a, b, the sequence U contains no alternating subsequence
$$a, b, a, b, \ldots$$
of length $s + 2$.

We write $U \in \mathrm{DS}(n, s)$.

Definition 2 (Extremal function)**.**

$$\lambda_s(n) = \max\{|U| : U \in \mathrm{DS}(n, s)\}. \tag{1}$$

2 Known results.

The first values are exact:

$$\lambda_1(n) = n, \qquad \lambda_2(n) = 2n - 1, \qquad \lambda_3(n) = 2n\alpha(n) + O(n), \tag{2}$$

where α denotes the inverse-Ackermann function. The fixed-order problem is now essentially settled. In particular,

$$\lambda_4(n) = \Theta\big(n2^{\alpha(n)}\big), \qquad \lambda_5(n) = \Theta\big(n\alpha(n)2^{\alpha(n)}\big), \tag{3}$$

For all n and s,

$$\lambda_s(n) \leq \binom{n}{2}(s + 1). \tag{4}$$

Moreover,

$$\frac{s}{\sqrt{n}} \longrightarrow \infty \quad \Longrightarrow \quad \lambda_s(n) \sim \binom{n}{2}s. \tag{5}$$

This does not determine the asymptotic behavior when $s = \Theta(\sqrt{n})$.

3 Problem formulation

For each fixed $c > 0$, determine the asymptotic behavior of

$$\lambda_{\lfloor c\sqrt{n} \rfloor}(n) \tag{6}$$

as $n \to \infty$.